Module 7
Reliability

Syllabus

Reliability: Basic concepts- Hazard function-Reliabilities of series and parallel systems-System Reliability - Maintainability-Preventive and repair maintenance-Availability.

Definition

The definitions given below with respect to a component hold good for a system or a device also.

If a component is put into operation at some specified time, say $t = 0$, and if T is the time until it fails or ceases to function properly, T is called the *life length* or *time to failure* of the component. Obviously $T (\leq 0)$ is a continuous random variable with some probability density function $f(t)$. Then the *reliability* or *reliability function* of the component at time 't', denoted by $R(t)$, is defined as

$$R(t) = P(T > t) = 1 - P(T \leq t) = 1 - F(t) \quad \text{where} \quad F(t) = \int_0^t f(t)\,dt = P(T \leq t)$$

$$\therefore \boxed{R(t) = \int_t^\infty f(t)\,dt}$$

By Fundamental theorem of calculus

$$\frac{dR(t)}{dt} = -f(t)$$

$$\Rightarrow \boxed{f(t) = -R'(t)}$$

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Hazard Function The instantaneous failure rate (or) Hazard function of the component is

$$\lambda(t) = \frac{f(t)}{R(t)} \quad \text{since} \quad f(t) = -R'(t)$$

$$\boxed{\lambda(t) = \frac{-R'(t)}{R(t)}}$$

Note:- From the definition of Hazard function

$$R(t) = e^{-\int_0^t \lambda(t)\,dt}$$

$$f(t) = \lambda(t)\, e^{-\int_0^t \lambda(t)\,dt}$$

Mean and Variance:-

Mean time to failure (MTTF) $= E(T) = \int_0^\infty t\, f(t)\, dt = \int_0^\infty R(t)\, dt$

$$\text{Var}(T) = \sigma_T^2 = \int_0^\infty t^2 f(t)\,dt - (MTTF)^2$$

Conditional Reliability

$$R(t\,|\,T_0) = P(T > T_0 + t \,|\, T > T_0) = \frac{R(T_0+t)}{R(T_0)}$$

$$= e^{-\int_{T_0}^{T_0+t} \lambda(t)\,dt}$$

Example:

The density function of the time to failure in years of the gizmos (for use

widgets) manufactured by a certain company is given by $f(t)=\frac{200}{(t+10)^3}, t \geq$ $t \geq 0$

(a) Derive the reliability function and determine the reliability for the fi first year of operation.

(b) Compute the MTTF.

(c) What is the design life for a reliability 0.95?

(d) Will a one-year burn-in period improve the reliability in part (a)? If so what is the new reliability?

Soln:- Given $f(t) = \frac{200}{(t+10)^3}$; $t \geq 0$

(a) $R(t) = \int_t^\infty f(t)\,dt = \int_t^\infty \frac{200}{(t+10)^3}\,dt = \left[-\frac{100}{(t+10)^2}\right]_t^\infty = \frac{100}{(t+10)^2}$

$R(1) = \frac{100}{(1+10)^2} = 0.8264$

(b) $MTTF = \int_0^\infty R(t)\,dt = \int_0^\infty \frac{100}{(t+10)^2}\,dt = \left(\frac{-100}{t+10}\right)_0^\infty = 10$ years.

(c) (Design life is the time to failure that corresponds to a specified reliability). In our problem take t_D is the design life.

To find t_D such that $R(t_D) = 0.95$

$$\frac{100}{(t_D+10)^2} = 0.95$$

$$\Rightarrow (t_D+10)^2 = 100.2632$$

$$\Rightarrow \boxed{t_D = 0.2598 \text{ years}}$$ (or) 95 days.

(d) $R(t \mid 1) = P(T > t+1 \mid T > 1)$

$$= \frac{R(1+t)}{R(1)} = \frac{\frac{100}{(t+1+10)^2}}{\frac{100}{11^2}} = \frac{121}{(t+11)^2}$$

If $R(t \mid 1) > R(t)$ then $\frac{121}{(t+11)^2} > \frac{100}{(t+10)^2}$

$$\Rightarrow \frac{121}{100} > \frac{(t+11)^2}{(t+10)^2}$$

$$\Rightarrow \frac{t+10}{t+11} > \frac{10}{11}$$

$$\Rightarrow 11t > 10t$$

we know $11t > 10t$ for any $t > 0$

$\Rightarrow$ one year burn in period will improve the Reliability.

$$R(1 \mid 1) = \frac{121}{(1+11)^2} = 0.8403 > 0.8264.$$

Example:

The time to failure in operating hours of a critical solid-state power unit has the hazard rate function $\lambda(t) = 0.003\left(\frac{t}{500}\right)^{0.5}$, for $t \geq 0$.

(a) What is the reliability if the power unit must operate continuously for 50 hours?

(b) Determine the design life if a reliability of 0.90 is desired.

(c) Compute the MTTF.

(d) Given that the unit has operated for 50 hours, what is the probability that it will survive a second 50 hours of operation?

Soln:- (a) $R(50) = \exp\left[-\int_0^t \lambda(t)\,dt\right] = \exp\left[-\int_0^{50} 0.003\left(\frac{t}{500}\right)^{0.5} dt\right]$

$= 0.9689.$

(b) $R(t_D) = 0.90$ we find t_D

$\exp\left[-\int_0^{t_D} 0.003\left(\frac{t}{500}\right)^{0.5} dt\right] = 0.90 \Rightarrow t_D = 111.54$ hrs.

(c) $MTTF = \int_0^\infty R(t)\,dt = \int_0^\infty \exp\left[-\int_0^t 0.003\left(\frac{t}{500}\right)^{0.5} dt\right] dt$

$= \int_0^\infty \exp\left[-\frac{0.003}{\sqrt{500}} \times \frac{2}{3} \times t^{3/2}\right] dt$

$= 451.65$ hrs.

(d) $P(T \geq 100 \mid T \geq 50) = P(T \geq \underbrace{50+50}_{100} \mid T \geq 50) = \frac{R(100)}{R(50)}$

$= \exp\left[-\int_{50} \lambda(t)\,dt\right] = 0.9438$

Example:

A manufacturer determines that, on the average, a television set is used 1.8 hours per day. A one-year warranty is offered on the picture tube having a MTTF of 2000 hours. If the distribution is exponential, what percentage of the tubes will fail during the warranty period?

Example:

A cutting tool wears out with a time-to-failure that is normally distributed. It is known that about 34.5% of the tools fail before 9 working days and about 78.8% fail before 12 working days.

(a) Compute the MTTF

(b) Determine its design life for a reliability of 0.99.

(c) Determine the probability that the cutting tool will last one more day given that it has been in-use for 5 days.